

Q1. List all the main components you find inside a desktop PC after opening the cabinet, and write a short description (1-2 lines) for each.

ANSWER ::

1. **Motherboard** – The main circuit board that connects all the computer parts together.
2. **CPU (Processor)** – The brain of the computer. It processes all instructions and calculations.
3. **RAM (Memory)** – Temporary memory used while programs are running. It helps the computer work faster.
4. **HDD (Hard Disk Drive)** – Stores the operating system, software, and files permanently.
5. **SSD (Solid State Drive)** – A faster storage device that helps the computer boot and load programs quickly.

6. **SMPS (Power Supply Unit)** – Converts AC power into DC power and supplies electricity to all components.
7. **CPU Cooler/Fan** – Keeps the processor cool and prevents overheating.
8. **GPU (Graphics Card)** – Displays graphics, videos, and games on the monitor.
9. **Heatsink** – Absorbs heat from the CPU or GPU and helps keep it cool.
10. **Cables** – Connect all the computer parts and carry power and data.

Q2. Take clear photos or draw labelled diagrams of each step as you disassemble a desktop PC, and organize them in a document with captions explaining what you did at each step.

Hint: Include steps like unplugging power, removing side panel, disconnecting cables, and removing RAM, HDD/SSD, and motherboard.

STEP 1 :: UNPLUG THE POWER



CAPTION ::

First, unplug the power cable from the desktop PC. This makes the computer safe to open.

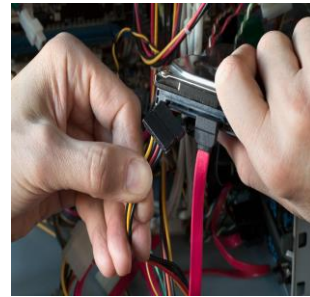
STEP 2 :: REMOVE THE SIDE PANEL



CAPTION ::

Remove the side panel of the PC cabinet using the screws. This allows you to see the internal components.

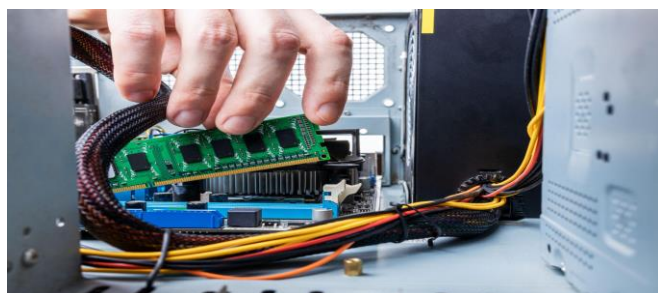
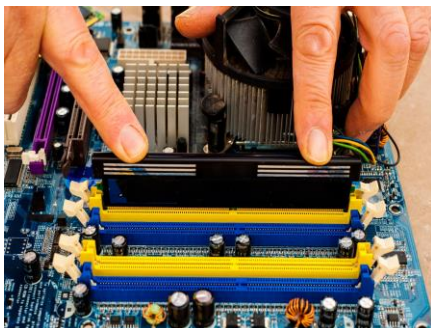
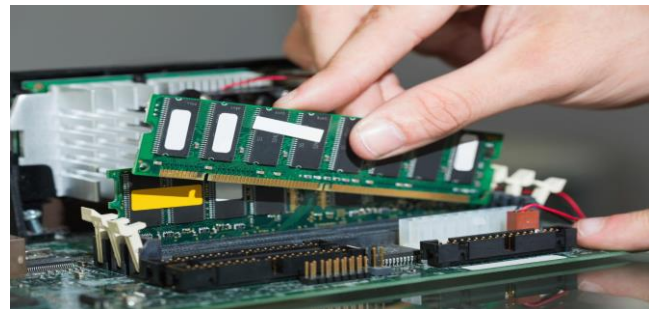
STEP 3 :: DISCONNECT INTERNAL CABLES



CAPTION ::

Carefully disconnect the power cables and data cables connected to the motherboard, HDD/SSD, and other components.

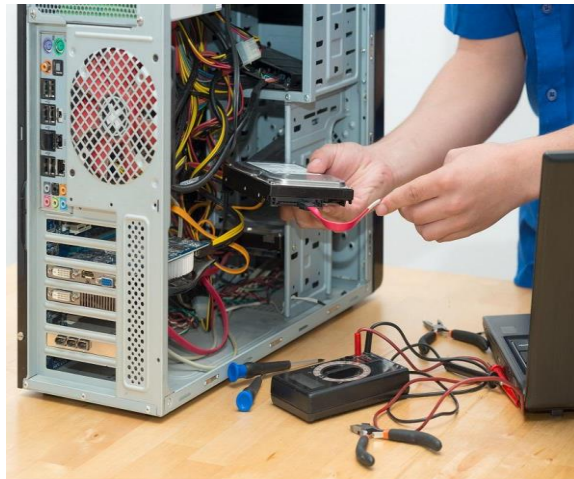
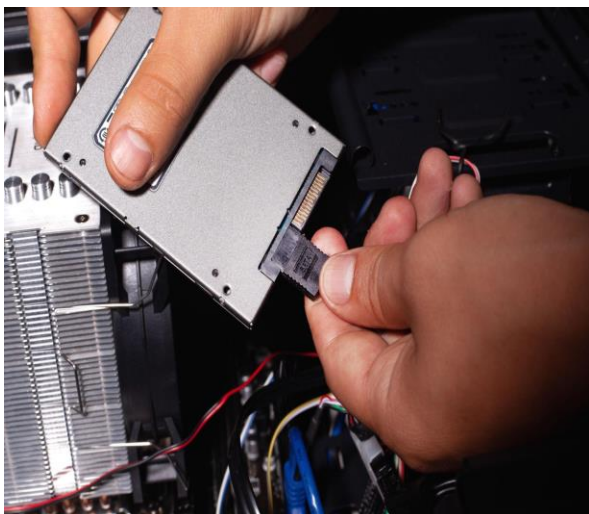
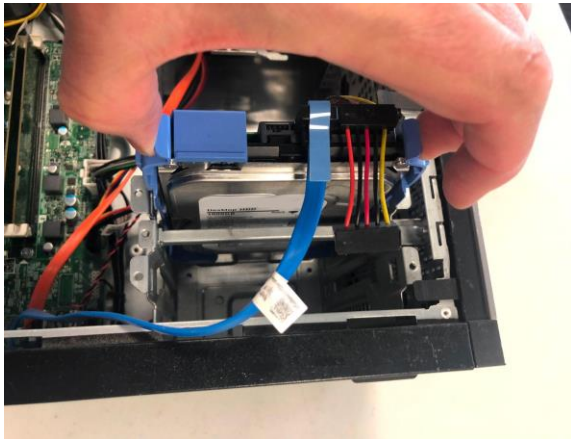
STEP 4 :: REMOVE THE RAM



CAPTION ::

Press the clips on both sides of the RAM slot and gently remove the RAM module.

STEP 5 :: REMOVE THE HDD AND SSD



CAPTION ::

Unscrew the HDD and SSD, disconnect their cables, and carefully take them out of the cabinet.

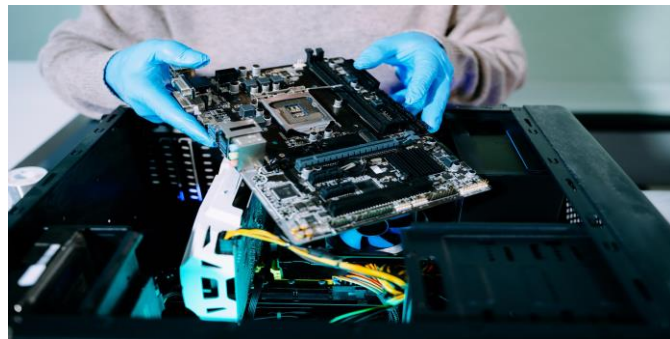
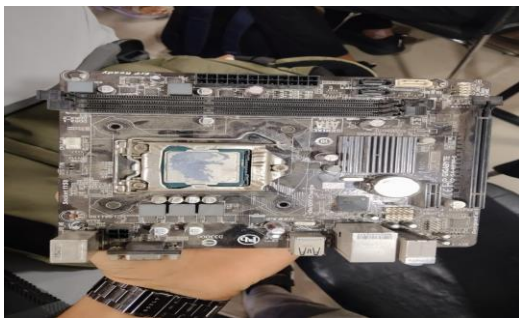
STEP 6 :: REMOVE THE GRAPHIC CARD (GPU)



CAPTION ::

Unscrew the graphics card, unlock the PCIe slot clip, and gently pull the graphics card out.

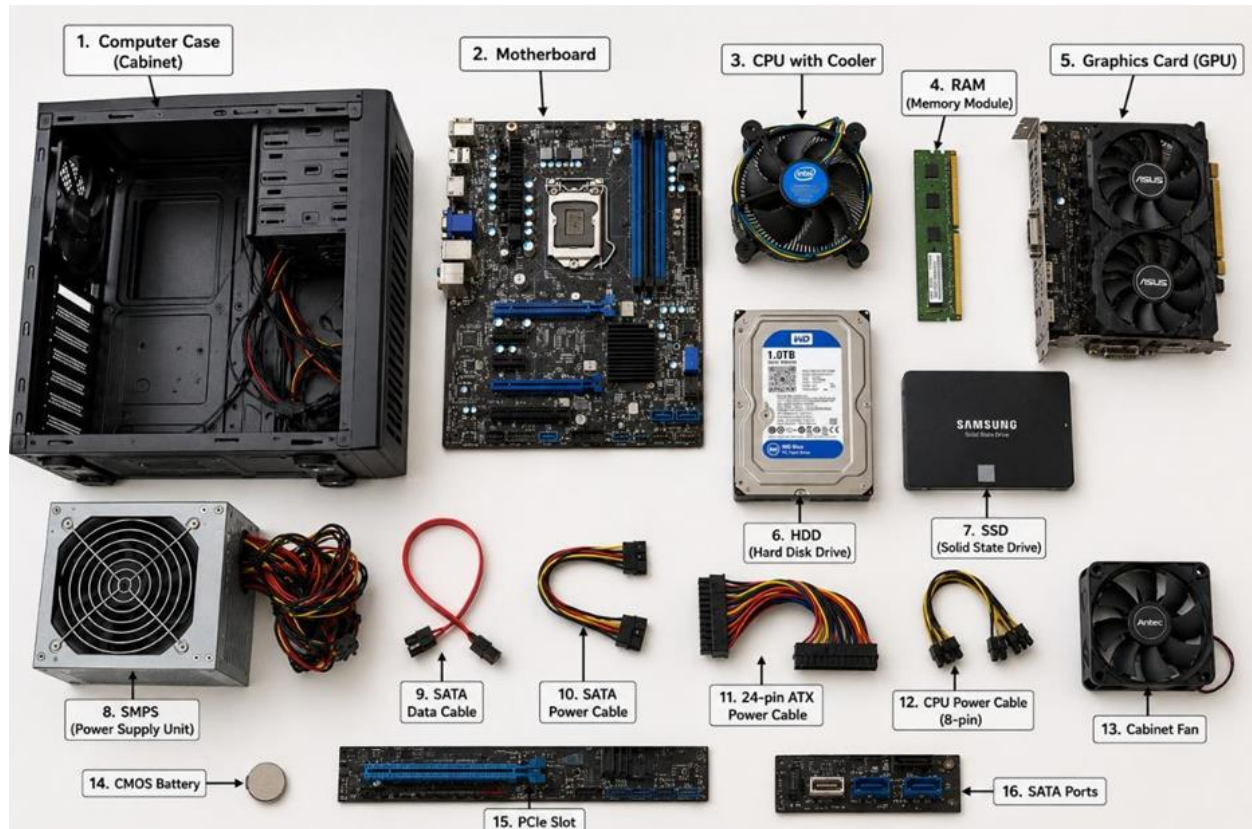
STEP :: 7 REMOVE THE MOTHERBOARD



CAPTION ::

Remove all screws holding the motherboard, then carefully lift it out of the PC case.

STEP :: 8 COMPONENTS ARE DISASSEMBLED



CAPTION ::

All the main components have been removed successfully. The desktop PC is now fully disassembled.

Q3. Reassemble the PC you disassembled, documenting each step with a brief explanation of what you connected or installed and why the order matters.

➤ **Step 1: Install the Motherboard**

Put the motherboard inside the PC case and tighten the screws.

Why: It is the main board that connects all the computer parts.

➤ **Step 2: Install the CPU**

Place the CPU in the CPU socket and lock it carefully.

Why: The CPU is the brain of the computer.

➤ **Step 3: Install the CPU Cooler**

Fix the CPU cooler on the CPU and connect the fan cable.

Why: It keeps the CPU cool and prevents overheating.

➤ **Step 4: Install the RAM**

Insert the RAM into the RAM slot until it clicks.

Why: RAM helps the computer run programs smoothly.

➤ **Step 5: Install the Power Supply (SMPS)**

Place the SMPS inside the cabinet and tighten the screws.

Why: It supplies power to all computer parts.

➤ **Step 6: Install the HDD/SSD**

Place the HDD or SSD in the drive bay and tighten the screws.

Why: It stores Windows, software, and files.

➤ **Step 7: Connect the Cables**

Connect the SATA cable, SATA power cable, 24-pin motherboard cable, and 8-pin CPU power cable.

Why: These cables provide power and transfer data.

➤ **Step 8: Install the Graphics Card (GPU)**

Insert the graphics card into the PCIe slot and tighten the screw.

Why: It improves graphics and gaming performance.

➤ **Step 9: Connect Front Panel Cables**

Connect the Power Switch, Reset Switch, USB, Audio, and Fan cables.

Why: These cables make the front buttons and ports work.

➤ **Step 10: Close the Cabinet**

Put the side panel back and tighten the screws.

Why: It protects the computer parts from dust and damage.

➤ **Step 11: Connect External Devices**

Connect the monitor, keyboard, mouse, and power cable.

Why: These devices are needed to use the computer.

➤ **Step 12: Turn On the PC**

Press the power button and check if the computer starts.

Why: It confirms that all the parts are connected correctly.

Q4. Create a checklist of safety precautions and tools needed before starting the PC assembly/disassembly process, and explain why each is important.

Turn Off the PC

Why: Prevents damage to the computer.

☐ **Unplug the Power Cable**

Why: Protects you from electric shock.

☐ **Press the Power Button (5–10 Seconds)**

Why: Removes any remaining power.

☐ **Work on a Clean and Dry Table**

Why: Keeps the computer safe from dust and water.

☐ **Wear an Anti-static Wrist Strap**

Why: Prevents static electricity from damaging the parts.

☐ **Hold Components by the Edges**

Why: Protects the chips and circuits.

☐ **Keep Screws in a Small Box**

Why: Stops screws from getting lost.

➤ **Tools Needed**

- ☐ **Phillips Screwdriver**

Why: Used to remove and tighten screws.

- ☐ **Anti-static Wrist Strap**

Why: Protects computer parts from static electricity.

- ☐ **Small Container**

Why: Keeps screws safe and organized.

- ☐ **Soft Brush**

Why: Removes dust from the computer.

- ☐ **Microfiber Cloth**

Why: Cleans the parts safely.

- ☐ **Cable Ties**

Why: Keeps cables neat and improves airflow.

- ☐ **Flashlight (Optional)**

Why: Helps you see small parts inside the PC